Practical problems — surface area and volume in context

The formulas are known; the whole skill is reading which one the situation wants.

LESSON 53–54 | 2 × 40 MIN | GEOMETRY 11, AMALIY MASALALAR | IGX 13.X IN CONTEXT

LESSON CLOCK

15' | 25' | 25' | 25' | 10'

Reading the question	15 min
Units	25 min
Density and rate	25 min
Which faces	25 min
Homework	10 min

LEARNING OBJECTIVES

- Choose between capacity, volume, curved area and total area from the wording.
- Convert between cm^3, litres and m^3 correctly in a single calculation.
- Use density and rate to turn a volume into a mass or a time.
- Report an answer to an accuracy the data supports.

TEXTBOOK

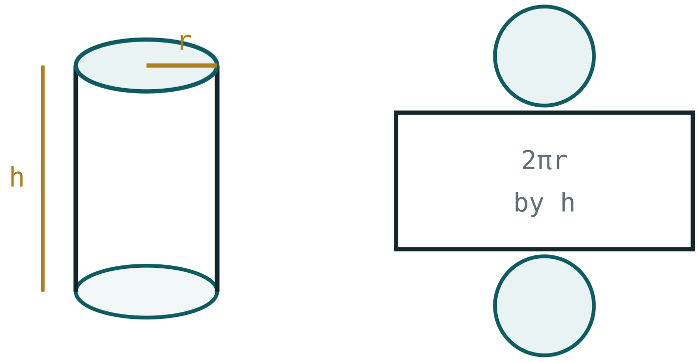
National	pp. 281–290
Cambridge	Core & Extended, pp. 379–388
Workbook	Exercise 13.6

01 Explanation

Reading the question

Nothing in Quarter IV is new. The whole difficulty is that the question no longer says which formula it wants — it describes a situation, and you must decide.

THE QUESTION SAYS	IT WANTS	WATCH FOR
“how much will it hold”	capacity — volume from **inner** dimensions	litres, not cm^3
“how much metal”	volume of material — a difference	outer minus inner
“how much paint”	surface area	which faces are painted
“how much label”	curved surface only	no circles
“how heavy”	volume × density	units of density
“how long to fill”	volume ÷ rate	rate units



$S = 2\pi r^2 + 2\pi rh$ — two circles and one rectangle

A cylinder unrolled — the label is the rectangle, the lid and base the two circles.

THREE QUESTIONS BEFORE ANY ARITHMETIC

- Am I asked for a **volume** or an **area**?
- Is the object **open or closed**, **solid or hollow**?
- What **units** does the answer want?

Answer all three in writing and the calculation itself is Grade 11 Quarter II work.

Units

$1 \text{ litre} = 1000 \text{ cm}^3$ $1 \text{ m}^3 = 1000 \text{ litres} = 1\,000\,000 \text{ cm}^3$

GIVEN IN	ANSWER WANTED IN	MULTIPLY BY
cm^3	*litres*	$\div 1000$
m^3	*litres*	$\times 1000$
m^3	cm^3	$\times 10^6$
m^2	cm^2	$\times 10^4$

CONVERT THE LENGTHS FIRST, NOT THE ANSWER

A trough 2.5 m long with a cross-section in cm has two different units in one formula. Convert 2.5 m to 250 cm **before** multiplying; converting the volume afterwards needs 10^6, and that is where the factor-of-a-thousand errors come from.

The squared and cubed conversion factors are the ones that catch people: $1\ m^2 = 10\ 000\ cm^2$ and $1\ m^3 = 1\ 000\ 000\ cm^3$, not 100 and 100.

Density and rate

$$mass = volume \times density \quad time = \frac{\text{volume}}{\text{rate}} \quad paint = \frac{\text{area}}{\text{coverage}}$$

QUANTITY	TYPICAL UNITS	EXAMPLE
density	g/cm^3	steel 7.8, water 1.0, aluminium 2.7
flow rate	$litres/min$	a tap 12 l/min
coverage	$m^2/litre$	paint 10 m²/l

Worked chain. A steel pipe, outer radius 9 cm, inner 7 cm, length 2 m:

STEP	WORKING	VALUE
length in cm	$2\ m = 200\ cm$	200
cross-section	$\pi(81 - 49)$	$32\pi\ cm^2$
volume	$32\pi \times 200$	$6400\pi \approx 20\ 106\ cm^3$
mass	$\times\ 7.8\ g/cm^3$	$\approx 157\ kg$

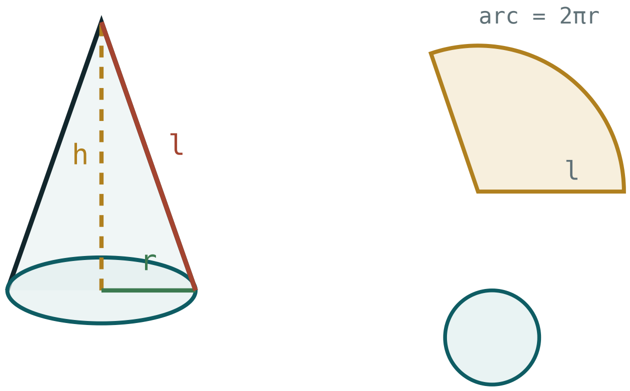
KEEP π UNTIL THE LAST LINE

6400π is exact; $20\ 106$ is not. Rounding at the volume stage and again at the mass stage compounds the error. One rounding, at the end.

Which faces

Surface-area questions turn on which faces exist and which are wanted.

OBJECT	AREA WANTED	FORMULA
a label round a tin	curved only	$2\pi rh$
a closed tin	total	$2\pi r^2 + 2\pi rh$
an open cylindrical tank	base + curved	$\pi r^2 + 2\pi rh$
a silo painted above ground	curved + dome	$2\pi rh + 2\pi r^2$
a bowl, inside and out	both curved surfaces	$2\pi(R^2 + r^2) + rim$
a conical tent with a floor	curved + base	$\pi rl + \pi r^2$



S = πr² + πrl — a circle and a sector

A cone unrolled — the sector is the curved surface, the circle the base.

THE COMMONEST SINGLE ERROR IN THE WHOLE QUARTER

Using $2\pi r^2 + 2\pi rh$ for an open tank. If the question says **open**, **tank**, **trough**, **bowl** or **tent without a floor**, at least one circle is missing. Read the noun, not just the numbers.

02 Worked examples

EXAMPLE 1 **A cylindrical tank of radius 1.5 m and height 4 m is open at the top. Find its capacity in litres and the area to be painted inside and out (not the base).**

1. $V = \pi(2.25)(4) = 9\pi \approx 28.27$ m³.
2. $= 28\ 270$ litres.
   $\times\ 1000$.
3. Curved area $2\pi(1.5)(4) = 12\pi \approx 37.7$ m², twice for inside and out.
4. ≈ 75.4 m².

ANSWER

$\approx 28\ 300$ litres; ≈ 75.4 m²

EXAMPLE 2 **A steel pipe has outer radius 9 cm, inner 7 cm and length 2 m. Find its mass at 7.8 g/cm³.**

1. Length 200 cm.
   Convert first.
2. $V = \pi(81 - 49)(200) = 6400\pi$
3. $\approx 20\ 106$ cm³.
4. $m \approx 156\ 800$ g ≈ 157 kg.

ANSWER

≈ 157 kg

EXAMPLE 3 **A tap delivering *12* litres a minute fills a cuboid tank $1.2 \times 0.8 \times 0.9$ m. How long does it take?**

1. $V = 0.864$ m³.
2. $= 864$ litres.
3. $t = \frac{864}{12} = 72$ minutes.
4. That is *1* h *12* min.

ANSWER

72 minutes

EXAMPLE 4 **A conical tent has base radius *3* m and slant height *5* m, with a groundsheet. Find the canvas needed and the air space inside.**

1. Curved $\pi rl = 15\pi \approx 47.1$ m².
2. Groundsheet $\pi r^2 = 9\pi \approx 28.3$ m².
3. Total $24\pi \approx 75.4$ m².
4. Height $\sqrt{25-9} = 4$; $V = \frac{1}{3}\pi(9)(4) = 12\pi \approx 37.7$ m³.

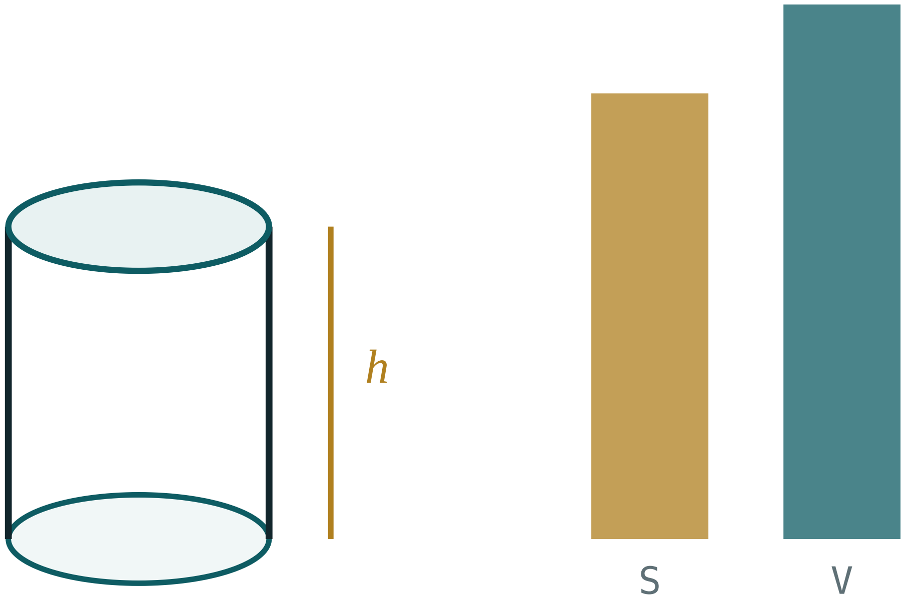
ANSWER

≈ 75.4 m² of material; ≈ 37.7 m³

03 Interactive model

Bring a tin, a bottle and a funnel; the class decides for each which formula the label, the contents and the metal need.

A cylinder, and what it holds



base area B **50.3** | perimeter P **25.1** | total S **251.3** | volume V **301.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Capacity	Sig‘im	Вместимость
Volume	Hajm	Объём
Density	Zichlik	Плотность
Flow rate	Oqim tezligi	Скорость потока
Wetted surface	Ho‘llanadigan sirt	Смачиваемая поверхность
Coverage	Sarf	Расход
Open solid	Ochiq jism	Открытое тело
Litre	Litr	Литр
Unit conversion	Birliklarni almashtirish	Перевод единиц
Degree of accuracy	Aniqlik darajasi	Степень точности

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

“How much will it hold” asks for:

- surface area
- capacity
- mass
- the curved area

1 m³ in litres:

- 100
- 1000
- $10\ 000$
- $1\ 000\ 000$

1 m² in cm²:

- 100
- 1000
- $10\ 000$
- $1\ 000\ 000$

An open cylindrical tank has area:

- $2\pi r^2 + 2\pi rh$
- $\pi r^2 + 2\pi rh$
- $2\pi rh$
- πr^2

A label on a tin needs:

- $2\pi rh$
- $2\pi r^2 + 2\pi rh$
- πr^2
- πrl

Mass equals:

- volume ÷ density
- volume × density
- area × density
- density ÷ volume

Rounding should happen:

- at each step
- once, at the end
- never
- twice

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Capacity of a cylinder $r = 2, h = 5$ cm, in cm³

   ANSWER $20\pi \approx 62.8$

2. Same, in litres

   ANSWER ≈ 0.0628

3. 1 m³ in litres

   ANSWER 1000

4. 2.5 m in cm

   ANSWER 250

5. Label area for $r = 3, h = 10$

   ANSWER $60\pi \approx 188$

6. Closed tin total area, $r = 3, h = 10$

   ANSWER $78\pi \approx 245$

7. Mass of 1000 cm³ of steel at 7.8 g/cm³

   ANSWER 7.8 kg

Medium · the standard question 7 PROBLEMS

1. Open tank $r = 1.5$, $h = 4$ m: capacity in litres

   ANSWER $\approx 28\ 300$

2. Same tank: area painted inside and out, base excluded

   ANSWER ≈ 75.4 m²

3. Pipe $R = 9$, $r = 7$ cm, $L = 2$ m: volume of metal

   ANSWER $6400\pi \approx 20\ 100$ cm³

4. Same pipe: mass at 7.8 g/cm³

   ANSWER ≈ 157 kg

5. Tank $1.2 \times 0.8 \times 0.9$ m at 12 l/min: filling time

   ANSWER 72 min

6. Conical tent $r = 3$, $l = 5$ with floor: material

   ANSWER $24\pi \approx 75.4$ m²

7. Same tent: air space

   ANSWER $12\pi \approx 37.7$ m³

Hard · stretch and reason

7 PROBLEMS

1. A silo, cylinder $r = 4$, $h = 10$ m with a hemispherical top: volume and painted area above ground

   ANSWER $\frac{608\pi}{3} \approx 637$ m³; $112\pi \approx 352$ m²

2. Paint at 10 m²/litre for that silo: litres needed

   ANSWER ≈ 36 litres

3. A hemispherical bowl of inner radius 12 cm: capacity in litres

   ANSWER ≈ 3.62 litres

4. A trough with trapezium section $40/60$ cm, depth 30 cm, length 3 m: capacity in litres

   ANSWER 450 litres

5. A gutter is half a cylinder of radius 8 cm and length 6 m: material area

   ANSWER 4800π cm² ≈ 1.51 m²

6. A tank of 2000 litres is a cylinder with $h = 2r$: find r

   ANSWER $r = \sqrt[3]{\frac{1}{\pi}} \approx 0.683$ m

7. A ball of radius 20 cm is painted at 2 so‘m per cm²: the cost

   ANSWER $3200\pi \approx 10\ 050$ so‘m

07 Homework

Homework — 5 tasks

Answer the three reading questions in writing before every calculation.

1. A cylindrical water tank is open at the top, radius 1.2 m, height 3 m. Find its capacity in litres and the area of metal used.
2. A copper pipe has outer radius 5 cm, inner 4 cm and length 3 m. Find the mass of copper at 8.9 g/cm³.

3. A tap delivers 15 litres a minute into a tank $1.5 \times 1.0 \times 1.2$ m. Find the time to fill it.
4. A conical tent of base radius 2.5 m and height 6 m has no groundsheet. Find the canvas needed and the volume inside.
5. For each of a tin label, an open trough and a solid steel ball, say which formula is needed and why, in one sentence each.